

VM14K Audit Journal

Purpose: Establish whether the dataset is usable, whether the proposed research question is workable, and what is worth reporting.

Sources:

- Project site: venera-ai.github.io/VM14K
- HuggingFace dataset repo: huggingface.co/datasets/venera-ai/VietnameseMedBench
- The three actual data files (data-processed-shuffled0/1/2.jsonl)
- Paper: <https://arxiv.org/pdf/2506.01305> "VM14K: First Vietnamese Medical Benchmark"
- GitHub repo: github.com/Venera-AI/VM14K-Megarepo

The question bank is real. 12,488 well-formed Vietnamese medical MCQs, all 34 specialties, downloads and parses cleanly. However:

1. **The released data is not what the paper describes.** There is no 4k/10k/2k split. That structure is mentioned *exactly once*, in the abstract, and never again in 14 pages. What exists is one 12,488-question pool (not 14k) released three times with the options shuffled (set 0 is the original, 1+2 are shuffled sets as in the answer options are shuffled). No held-out set or leaderboard provided.
2. **The data is neither deduplicated nor internally consistent.** Even though the paper says the 14,000 were sampled from a deduplicated pool (§3.2.2) and that all 14,000 were *verified* (§6). **The pairs found are in the .md file. 25 groups sampled from the 207-group loose-flag list. 9 were genuine key contradictions; the rest were other pipeline defects**
3. a VM14K.jsonl exists in the repo with 10,000 rows but only 5,435 unique IDs, not existing on hugging face- purpose unclear.

1. Data quality

Everything here was re-measured from **data-processed-shuffled0.jsonl** (the original unshuffled dataset) before being written down.

1.1 The advertised release structure does not exist

The paper promises a sample set, a full set, and a private leaderboard set. None of that is in the release.

The three files each contain **12,488 rows** with the **same question IDs in the same row order**. They differ only in the order of the answer options. 12,488 matches none of 4,000 / 10,000 / 14,000, nor any sum of them.

Difficulty levels: all 4 tiers present, heavily imbalanced

Easy 4,112 / Medium 7,093 / Challenging 1,193 / Hard 90 (of 12,488). Hard = 0.7% of the data. The paper states "Challenging and Hard questions are combined under 'Hard'" specifically for its results tables (Tables 2–3) likely because Hard alone is only 0.7% of the data.

The evaluation notebook loads internal batches named **final-data-7k** and **final-data-5k3** ($\approx 7,000 + 5,300$) and scores against the combination.

Where did the other 1,512 questions go?

12,488 released vs 14,000 claimed. We cannot account for the gap.

Deduplication cannot explain it, dedup only shrinks a set, and 12,488 is already *below* 14,000. Withholding a 2,000-question private set would give 12,000, not 12,488.

1.2 Duplicates and contradictions (CHECK DOC: VM14K_manual_verification)

§3.2.2 says the 14,000 were sampled **from a deduplicated pool**. §6 (Limitations) says they were “able to **verify** 14,000 multiple-choice questions within 3 months.” So the released file is presented as post-dedup *and* post-human-verification. However:

19.6% of released rows are duplicate questions

Grouping by (normalised question + normalised option set) using **the authors' own normaliser** from **dedup_utils.normalize_vietnamese**: **1,125 duplicate groups covering 2,450 rows (19.6%)**.

Group sizes run from 2 up to 6 copies of the same question.

The difficulty labeller disagrees with itself

Difficulty was LLM-assigned (paper Table 1), so duplicates give a free test–retest reliability check on the labeller.

- The paper says: “For the questions that do not have difficultLevel and medicalTopic, we used GPT-4o and Geimini Flash to label those based on our definition”

The labeller get confused in both directions, including Challenging↔Hard (already small set) - as shown in the manual check, can be explored further for the severity.

```
### Group 62 – topic: Endocrinology, Gastroenterology
```

```
Q: Nhóm cơ chất nào sau đây có thể được sử dụng để tổng hợp glucose?
```

```
line 1944 | Challenging | id 5fd1e29e4fd348fa825f6bd40994ddd3 | ANSWER: Glycin, glycertol, mannose
```

```
line 6176 | Easy | id b618386cbdc2482799c7a2537341474c | ANSWER: Acid palmitic, acid aspartic, glycerol
```

1.3 Format problems

A number of the questions in the dataset is not a 4-option MCQ.

Line numbers and count for the exceptions:

2-option: 1,240 lines (~10% of the dataset) have exactly 2 options, these are Đúng/Sai (True/False) questions, not 4-option MCQ. This is by far the largest non-4-option chunk in the file: over 60x the size of everything else combined. Most are literally ["Đúng", "Sai"] (or lowercase/with punctuation/Đ/S variants, 235 of the 1240 have minor formatting variants like ['Đ', 'S'] or ['1/10.000', 'Sai'], but all are still binary).

1-option (15): 4516, 4528, 4529, 5422, 5423, 7188, 7893, 8133, 8620, 8769, 9417, 11224, 11236, 11237, 12485 -> models get a free “correct” on these

- E.g. (4516): {"id": "fd6d1063f1614cba9c18bfac8fe57d5e", "difficulty_level": "Easy", "medical_topic": ["Toxicology", "Environmental Health"], "question": "Furadan là thuốc trừ sâu thuộc loại:", "options": ["Thuốc diệt côn trùng dị vòng carbamat"], "option_map": [0], "answer": "A", "answer_index": 0}

Effectively 1-option / unanswerable (21): the above plus 8946, 10264, 10265, 10266, 10267, 10809

- E.g. (8946): {"id": "03c9a411d2244c1c937ed64e0bflc10a", "difficulty_level": "Challenging", "medical_topic": ["Pulmonology", "Intensive Care", "Emergency Medicine"], "question": "Trong hội chứng suy hô hấp người lớn (ARDS) những phát hiện nào sau đây có mặt, ngoại trừ:", "options": ["Tăng áp mao mạch phổi", "optionB", "optionC", "optionD"], "option_map": [0, 1, 2, 3], "answer": "A", "answer_index": 0}

3-option (101): 90, 109, 269, 421, 422, 697, 798, 804, 805, 807, 1035, 1040, 1042, 1043, 1187, etc.

- E.g. (90): {"id": "4235409707e84f7ca76af6891a7ba20c", "difficulty_level": "Easy", "medical_topic": ["Urology", "Infectious Diseases"], "question": "Thời gian tinh hoàn trong bệnh quai bị hết sưng trở lại bình thường:", "options": ["7-9 ngày", "10-15 ngày", "7-10 ngày"], "option_map": [0, 1, 2], "answer": "C", "answer_index": 2}

5-option: 8188, 8821

- Note line 8188's correct answer is "Câu a, b đúng" ("a and b are correct"). **Shuffling breaks it since it changes what is A and what is B. => Other answers like “A and C are correct”, “C and D are correct” would be affected!**
- E.g. (8188): {"id": "428685a07cc04e49ba09e36f7a5a2470", "difficulty_level": "Medium", "medical_topic": ["Orthopedics", "Endocrinology", "Obstetrics and Gynecology"], "question": "Estrogen ngăn chặn hoặc trì hoãn sự tiêu xương của phụ nữ sau mãn kinh là do:", "options": ["Kích thích sản xuất Calcitonin.", "Ức chế tác dụng của PTH.", "Kích thích thay thế các xương đã mất.", "Làm tăng khoáng hóa"], "option_map": [0, 1, 2, 3, 4], "answer": "A", "answer_index": 0}

```
xương.", "Câu a, b đúng."], "option_map": [0, 1, 2, 3, 4], "answer": "E",  
"answer_index": 4}
```

7-option (19 (has **optionE `optionF` `optionG` etc. as options)):** 7866, 8099, 8126, 8302, 8399, 8729, 8730, 8785, 8808, 8826, 8833, 8861, 8888, 8934, 8961, 8996, 9007, 9011, 9063

- **All have literal optionE/optionF/optionG placeholder junk padding real 4-option questions.**
- E.g. (7866): {"id": "f05f7fb9cae34c03a7091e5298944505", "difficulty_level": "Challenging", "medical_topic": ["Obstetrics and Gynecology", "Pharmacology"], "question": "Câu nào sau đây sai khi nói về phụ nữ có thai khi so với người bình thường?", "options": ["Sự hấp thu ở vùng mông không đều", "Sự hấp thu ở vai tăng", "Giảm áp lực thủy tĩnh ở hệ tĩnh mạch (tăng)", "Sự tưới máu vào mô ngoại biên tăng", "optionE", "optionF", "optionG"], "option_map": [0, 1, 2, 3, 4, 5, 6], "answer": "C", "answer_index": 2}

4-option containing junk (15): 2886, 3878, 5002, 8774, 8946, 9755, 9904, 9966, 10170, 10264, 10265, 10266, 10267, 10809, 12094

- E.g. (2886): {"id": "84bd11600f564b2ca0ee57edf6827acd", "difficulty_level": "Medium", "medical_topic": ["Obstetrics and Gynecology"], "question": "Triple test định lượng để phát hiện 2 bệnh lý bất thường nào của thai:", "options": ["Tật nứt cột sống", "Bất thường về nhiễm sắc thể", "Học 2 cái này", "optionD"], "option_map": [0, 1, 2, 3], "answer": "C", "answer_index": 2}

The paper reports a single aggregate accuracy across all of these. => This influences the leaderboard.

The topic field is built via a raw row["medicalTopic"].split(",") with zero validation.

The paper claims **34 specialties**. The data contains **133 distinct topic strings**.

- The paper says: "For the questions that do not have difficultyLevel and medicalTopic, we used GPT-4o and Gemini Flash to label those based on our definition"

The extras include a literal **optionD**; a whole Vietnamese sentence stored as a topic (**lí giải minh bạch**); and comma-split wreckage like **"Endocrinology / Internal Medicine"**.

- Line 11912: {"id": "214ddf96c58c46928e2a32022c3d40e0", "difficulty_level": "Easy", "medical_topic": ["lí giải minh bạch"], "question": "Tiêu chuẩn Nugent cho chẩn đoán loạn khuẩn âm đạo có những đặc điểm gì?", "options": ["Dùng như tiêu chuẩn vàng", "khó thực hiện", "lí giải minh bạch", "dễ thực hiện"], "option_map": [0, 1, 2, 3], "answer": "A", "answer_index": 0}

The 133-count is further inflated by near-duplicate canonical-looking labels that aren't typos/junk but definitely aren't the same string — Cell Biology vs Cellular Biology, Infectious Diseases vs Infection Diseases, Intensive Care vs Intensive Care Medicine, Pain Medicine vs Pain Management, plus two empty-string (") topics.

Related: 20 rows are tagged **Other(No Category)**, which **deduplication.py** is supposed to filter out. But that filter is a string-equality test on the whole cell, so a row tagged **["Gastroenterology", "Other(No Category)"]** never matches and survives. **The filter is broken.**

△ Normalise medical_topic against the official 34 before ANY specialty-level analysis. About 99 non-canonical values need dropping or re-mapping.

Extraction artifacts left in the data

9 rows have options that ran together during extraction e.g. an option reading **"Cạp vào niêm mạc ruột vật chủ và hút máu B. Thủy phân mô vật"**, where the "B." marker and half of option B were swallowed into option A.

- E.g. (9364): {"id": "90a227237a0443bf8e694de34b2af6bc", "difficulty_level": "Easy", "medical_topic": ["Parasitology"], "question": "Hình thức dinh dưỡng của giun tóc (Trichuris trichiura):", "options": ["**Cạp vào niêm mạc ruột vật chủ và hút máu B. Thủy phân mô vật**", "**chủ và hút máu**", "**Cắm đầu vào màng nhầy ruột để hút chất dinh dưỡng D. Ăn chất**", "**cặn bã của cơ thể ký chủ**"], "option_map": [0, 1, 2, 3], "answer": "A", "answer_index": 0}

1 row contains the literal sentinel **INVALID_OPTION** as option text.

- E.g. (596): {"id": "75bf0cf82ff443cba4b4ed9bf7356432", "difficulty_level": "Easy", "medical_topic": ["Gastroenterology", "Infectious Diseases"], "question": "Chọn một câu trả lời đúng. Phát biểu nào là đúng khi đề cập đến bệnh viêm gan vi rút:", "options": ["**Rượu, bia, lao động quá sức... là những yếu tố làm bệnh viêm gan nặng lên** Đã có vaccin phòng bệnh viêm gan A và C.", "**Bệnh nhân viêm gan vi rút sau khi ra viện cần ăn kiêng (thịt, mỡ..) 6 - 12 tháng tùy theo mức độ bệnh.**", "**Theo dõi tình trạng tắc mật không cần theo dõi các xét nghiệm chỉ cần theo dõi các dấu hiệu trên lâm sàng.**", "**INVALID_OPTION**"], "option_map": [0, 1, 2, 3], "answer": "A", "answer_index": 0}

1.4 The shuffle

The shuffle mechanic itself is flawless

Across $12,488 \times 3$ files: **answer_index** always points to the same option text, and the answer letter always matches the index. **Zero errors.**

Also confirmed: **shuffled0** is **not shuffled**.

DataCleaning/final_transform.ipynb builds one full "rows" list from cleaned source data, writes it once to data-processed-shuffled0.jsonl, then calls a shuffle_options() function on that same list twice more to produce shuffled1 and shuffled2 only randomizing answer-choice order and

recomputing answer/answer_index. There is no subsetting, sampling, or train/test logic anywhere in this notebook.

The source answer key is positionally biased

In the unshuffled source order, the correct answer is **A 28.7%** / B 24.1% / C 22.5% / D 24.8%.

With 2 options, a "pick A" prior gives you **53.8%** instead of 50%. That's nearly 4 free points on 1,240 questions. Small in absolute terms, but it's a systematic freebie, and it favors weak models (who lean on position priors) over strong ones (who read the content). Same direction as the coin-flip problem: it compresses the leaderboard.

Shuffling flattens it to ~25% each. So the shuffle-ensemble is silently correcting a real defect in the source key — one the paper never reports. Any single-shot evaluation on file 0 rewards a model's "pick A" prior.

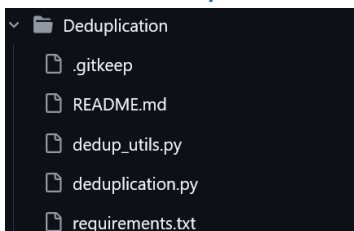
Another concern: Shuffling corrupts options that refer to other options

Questions contain an option whose text is a **positional reference** “Cả A và B” (both A and B), “Câu A và C đều đúng”.

When you shuffle, the option *text* moves but the letters it names now point at different content.

The paper's ensemble metric runs models on shuffled orders and votes. So is its headline metric computed partly on items the shuffle broke?

2. The deduplication



- There was a deduplication process, then why is the released data dirty? (section 1)
- Could the data released be the dataset BEFORE deduplication? But the paper says otherwise. Dedup happened -> THEN we got the dataset (which is not exactly 14k but a bit smaller)
 - “Additionally, *removal based on clustering and the Levenshtein distance* is suitable [...] Afterwards, the answer verification step includes three sources of references: (1) the correct answers from the source data, (2) the answers from foundation LLMs, and (3) the answers from medical experts[...] *we sampled 14,000 questions from a total of 100,000*”

Question: What was the human evaluation process like? There is no human-answer, source-answer or LLM-answer column/cataloguing for the “correct answers” in the dataset.

- **POSSIBLE REASON IN `final_transformation`:**
`TfidfVectorizer(min_df=5, max_df=0.8)` is fit **on the question text only** — options are never vectorized. Any question whose words all fall below `min_df=5` becomes a zero vector, cosine distance 1.0, and is **never even proposed as a duplicate candidate**.

3. I briefly checked [benchmark_analysis.ipynb](#)

Appendix B of the paper says “*We used the same prompt for all models*” and shows one prompt ending in “Only response with 1 character.” The paper describes nothing else about how answers were extracted.

Model	Cell	How the answer is extracted
GPT-4o, o3-mini	17	<code>json.loads(content)["answer"]</code>
Llama-4 Maverick	28	<code>json.loads(content)["answer"]</code>
Gemini 2.0 Flash	49	<code>content["answer"]</code> (already a dict)
DeepSeek-R1	54	strips <code><answer>...</answer></code> tags
Qwen3-32B	43	<code>answer0.strip()[-1]</code>

Qwen3's `.strip()[-1]` doesn't have a safety net:

- Hypothesis: If the last character happens to be a valid option letter, but isn't the model's actual answer.
- E.g. a reasoning model like Qwen3 (which is explicitly a chain-of-thought model) writing something like: “...loại trừ C, đáp án đúng là B, không phải A” → last char is “A” → valid letter → passes straight through → recorded as A when the model actually said B.

For GPT/o3-mini/Llama/Gemini/DeepSeek, “extraction failed” and “model answered wrong” are cleanly separable — a parse failure just removes a row from the average rather than scoring it.

For Qwen3, some unknown fraction of what's being scored as “wrong” could be the extractor picking the wrong letter out of correct reasoning or worse: some unknown fraction of what's scored as “right” could be a coincidental letter match that isn't really Qwen3's stated answer. That makes Qwen3's row in the leaderboard not apples-to-apples with the other four: its accuracy figure is a mix of “model accuracy” and “extractor accuracy,” while the others are just “model accuracy.”

EXPERIMENT: run Qwen3-32B on the public set via vLLM with the paper's exact prompt. Apply `output.strip()[-1]`, check the concern.

NEXT STEP: try reproducing their pipeline from the github code.

- **Problem? The evaluation notebook references paths like**
`"../<placeholder>-interim/file_manual_v2/final-data-7k"` — a literal, non-functional

placeholder string appearing multiple times. This suggests an internal directory or company name was redacted before publishing but never cleaned into a runnable path; the notebook would not execute as-is for someone cloning the repo.

- exact count of shuffle-breaking position-referential answers
- exact count of image-dependent questions
- frequency of malformed options
- estimate of how much benchmark accuracy changes after cleaning
- reproduce at least one published result on the released data

Worth evaluating: The “Vietnamese capability” conclusion

The paper's central surprise (§4.1): reasoning models “are not better than non-reasoning models”, “in direct conflict” with their English MedQA results. Their explanation: **“other factors, such as Vietnamese capability, are better predictors of performance than medical ability in English.”**

Is the original research question viable?

The proposal was: *train a local Vietnamese medical model on this dataset, then compare it against frontier models.*

- There is no train/test split, so we would have to build one, and this dataset is dirty - dedup needed. **Training on a dirty set and evaluating on the same dirty set produces a number nobody can interpret.**
- The paper already has 17 models in its table.
- It is the most expensive path: GPU time, hyperparameter search, weeks of iteration and at the end, one number that may be an artifact of the problems above.

Other possible options?

Benchmark audit + cleaned release

Deliverable: a paper documenting the data-quality and harness problems, revise the dataset to get a deduplicated, contradiction-flagged, corruption-quarantined version of the benchmark, with the models re-scored on it.

The general methodological findings

1. **Option-order shuffling silently corrupts position-referential MCQ options.** “All of the above” breaks when you shuffle. This affects *any* benchmark using shuffle-debiasing
2. **Natural duplicates give you a free reliability measurement on an LLM labeller.** We measured the difficulty labeller disagreeing with itself on identical items 19.8% of the time.

3. ...

Could be a short methods paper with VM14K as the case study.